1 Introduction

1.1 Purpose

The purpose of this document is to provide the software requirement specifications for the program Gold Star.

1.2 Scope

ADHD causes a lot of challenges in terms of building and following routines. The ways in which people experience ADHD are varied and as such a lot of possible solutions don't work for people with it. The program Gold Star will be a system to build and follow routines that is flexible for all kinds of ADHD.

1.3 Definitions, Acronyms, and Abbreviations

Activity Alternation – Going from an activity requiring high mental focus to an activity requiring low mental focus.

Activity Switching – The cognitive process of disengaging from one task and shifting mental focus to another.

Adaptive Scheduling – To schedule a task which can be completed anytime within a set timeframe. For example, a task which can be completed anywhere inclusively between 7:00 P.M. and 10:00 P.M.

Application Programming Interface (API) - A set of standard rules, protocols, and definitions allowing different software applications to interact with one another.

Attention-Deficit/Hyperactivity Disorder (ADHD) - A common neurodevelopmental disorder that alters the brain's ability to plan, focus, remember instructions, and manage emotions.

Cascading Style Sheets (CSS) - A style sheet language used to determine the visual presentation, layout, and aesthetics of a website.

Git - A digital version control system intended to preserve a history of saved modifications, additions, and deletions to software source code.

GitHub - A cloud-based platform where developers can collaborate on software code.

Gold Star - A web-based habit-building and time management platform designed specifically for individuals with ADHD.

Habit Stacking – Interspersing new habits with existing habits to make task completion easier to complete without becoming overwhelmed.

Hyperactivity - A state of being unusually or abnormally active.

HyperText markup Language (HTML) - The standard markup language used to create and structure webpages.

Javascript (JS) - A lightweight programming language used to build dynamic interactivity on web pages.

Jest - An open-source Javascript testing framework.

JSDoc - An API documentation generator and syntax standard for JavaScript.

Pomodoro Technique – A time-management technique which separates work into 25-minute increments and 5-minute-long breaks. After repeating this technique 4 times, a longer break from 20 to 30 minutes is taken before the process repeats.

PostgreSQL - An open-source RDBMS that stores, manages, and retrieves data.

Reinforce - To make something stronger, more effective, or more compelling by adding support.

Relation Database Management System (RBMS) - software used to store, manage, query, and retrieve data structured in table form.

Rigid Scheduling - To schedule a high-priority task which must either be initiated at or completed before an established time. For example, a task which must start at 6:00 P.M. or must be completed by 6:00 P.M.

Routine Adaptation – A requirement for an individual to change one's schedule or create new routines.

Routine - A sequence of actions regularly followed, or a fixed program.

Task Deviation – When a task has an unexpected change regarding completion. A task may be accomplished out of order in one's schedule, or it may not be accomplished at all.

Visual Studio Code (VSCode) - A development environment used for software engineering and web development.

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1.5 Overview

Section 2 goes into detail about what the program will be overall. Section 3 will go into detail about all the requirements involved with the system.

2 Overall Description

2.1 Product Perspective

Gold Star will be a web-based program for building habits and routines. It evolves with the user's ability to complete tasks to lessen the feeling of failure that comes with ADHD when struggling with tasks.

2.2 Product Functions

The prototype will have a calendar view that lets the user see all their tasks at once, adaptive scheduling that tailors tasks to what level of activity the user is okay with, task prioritization so that tasks that are more necessary can be done on time, and finally habit stacking to help integrate new tasks into the user's routine. In the finalized build, the ui will also be customizable and there will be proper rewards for completing tasks.

2.3 User Characteristics

There will only be one class of user, people with ADHD or those who struggle in similar ways. The users will use the program individually with no interaction or other classes of user.

2.4 Constraints

N/A

2.5 Assumptions and Dependencies

N/A